The Missing Star: Measuring the Natural Rate of Exchange*

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Abstract

This paper contributes quarterly time-varying estimates of the natural rate of exchange (q_t^*) for the United States from 1970Q1 to 2025Q4, defined as the real effective exchange rate consistent with output at potential, stable inflation, and a stable trade balance. A semi-structural open-economy state-space model captures the role of the real exchange rate in macroeconomic dynamics and is estimated by a multi-stage maximum likelihood procedure. Misalignments between the real effective exchange rate and the time-varying natural rate of exchange are substantial and persistent over the historical sample, with out-of-sample forecasts outperforming a random walk at horizons of up to four years.

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Keywords: natural rate of exchange, equilibrium real exchange rate, state-space models

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1 Introduction

The real exchange rate is among the central macroeconomic quantities for which no standard measure of equilibrium is in routine use. Contrastingly, well established measures of potential output and the natural rates of unemployment and interest are widely used in macroeconomic analysis, with estimates reported alongside measures of uncertainty and revised as new data is accumulated. Comparable time-varying estimates of the real exchange rate consistent with internal and external stability are scarce within the existing literature, which focuses largely on estimating the level of equilibrium rather than its evolution over time.

The lack of time-varying estimates of the equilibrium real exchange rate matters because claims of currency misalignment cannot otherwise be meaningfully evaluated. In particular, if the natural rate of exchange changes materially across time, an estimate of its level alone provides an unreliable reference point for judging the pressures the real exchange rate places on the domestic economy. Treating the equilibrium as a latent state within a semi-structural model allows its path to be estimated jointly with the relationships that define it. Variation in the real exchange rate can then be decomposed into shifts in the equilibrium and departures from it, allowing the size and persistence of misalignment to be quantified.

Existing estimates of the equilibrium real exchange rate fall into three broad categories, namely purchasing power parity, the behavioural equilibrium exchange rate and the macroeconomic balance approach (Ca' Zorzi et al., 2022). In the theory of purchasing power parity, the equilibrium real exchange rate is considered constant, measured by the long-run mean of the real exchange rate, deviations from which are persistent (Rogoff, 1996; Taylor and Taylor, 2004). Behavioural equilibrium exchange rate approaches allow for variation, regressing the real exchange rate on fundamentals such as net foreign assets, the terms of trade and relative productivity (Clark and MacDonald, 1999; Lane and Milesi-Ferretti, 2004; Ricci et al., 2013). The macroeconomic balance approach defines the equilibrium as the real rate consistent with output at potential and a current account at a norm estimated from medium-term fundamentals (Chinn and Prasad, 2003; Lee et al., 2008).

In the latter two approaches misalignment is understood as the part of the real exchange rate that observed fundamentals leave unexplained. Such fundamentals, however, account for only part of the movement in the exchange rate (Meese and Rogoff, 1983; Cheung et al., 2005; Itskhoki and Mukhin, 2021), which has also been linked to financial forces, such as risk premia and the liquidity value of government bonds, that are difficult to measure over long samples (Gabaix and Maggiori, 2015; Engel, 2016). In contrast, pure statistical time-series decompositions rest at the other extreme, separating permanent from transitory movements without the imposition of any macroeconomic structure. More recent unobserved-components models

(Holston et al., 2017) occupy a desirable middle ground, extracting the equilibrium from the joint behaviour of macroeconomic variables; retaining a degree of theoretical structure while allowing the equilibrium to evolve as a latent trend, so that it is neither fixed nor confined to the observed fundamentals.

This paper extends a decomposition of this nature to the real exchange rate. The natural rate of exchange (q_t^* or q-star) is defined as the real exchange rate consistent with output at potential, stable inflation, and a stable trade balance. q_t^* is estimated for the United States from 1970Q1 to 2025Q4 on a quarterly basis, using a semi-structural open-economy model in four observables, namely output, inflation, the real effective exchange rate and net exports. The real exchange rate gap operates through three core channels, influencing demand in the dynamic IS curve, prices through pass-through in the Phillips curve and finally net exports through expenditure switching in the trade balance. The division between the natural rate and the gap rests on a median-unbiased estimate of the natural rate's volatility. The natural rate of exchange combines a latent trend with an observed component in the terms of trade, reflecting the close link between commodity prices and the real exchange rate, including that of the dollar (Chen and Rogoff, 2003; Ayres et al., 2020).

The closest application to the exchange rate is by Berger and Kempa (2014), who estimate the equilibrium real exchange rate for Canada jointly with potential output and the natural rate of interest. In their model, the equilibrium is a random walk and the exchange rate gap follows its own autoregressive process, with priors used to discipline the volatility of the trend. This paper differs in several dimensions. The natural rate of exchange is defined by internal and external balance, with net exports entering as an observable, and part of it is tied to the observable terms of trade. The exchange rate gap responds to the differential between home and foreign long-term yields, so that the contribution of interest parity to misalignment can be measured. The volatility of the natural rate is identified by median-unbiased estimation rather than priors, and one-sided estimates are reported alongside two-sided ones, such that the reliability of misalignment at the time can also be assessed.

Several key findings emerge from this empirical analysis. Misalignments between the real exchange rate and its natural rate are substantial, reaching roughly 20 percent at the dollar peak of the mid-1980s and negative 20 percent over the late 1970s. The exchange rate gap is also highly persistent, maintaining the same sign for approximately 4 years on average. The natural rate of exchange itself declines by 33 percent to a low in 2008Q3 and increases by 19 percent towards end of the sample, with the observed terms of trade component accounting for more than 50 percent of its quarterly variation. Finally, pseudo out-of-sample forecasts of the real exchange rate outperform a random walk at horizons of one to four years, reducing the mean squared error by up to 31 percent.

The remainder of this paper is structured as follows. Section 2 sets out the semi-structural model, Section 3 discusses the implementation, Section 4 presents and discusses estimates of the natural rate of exchange, dollar misalignment, and out-of-sample forecasts. Section 5 focuses on robustness. Finally, Section 6 concludes. Sections A and B outline the data and the full state-space specification of the model at each stage respectively.

2 Framework

2.1 Theoretical background

The equations estimated below are reduced forms of the canonical open-economy New Keynesian structure, together with a standard model of trade volumes in which home and foreign goods are imperfect substitutes.

In the open-economy frameworks of Clarida et al. (2002), Galí and Monacelli (2005) and Monacelli (2005), log-linearisation about a symmetric steady state yields three key relations: an aggregate demand equation in which the output gap depends on its expected future value, the real interest rate gap and the real exchange rate, the latter entering through expenditure switching; a Phillips curve in which inflation depends on expected inflation, the output gap and the real exchange rate, via import prices; and an interest rate parity condition relating the expected movement in the real exchange rate to the real interest differential.

Replacing expected future values with their lagged counterparts, as is standard in semi-structural measurement models of this class, yields backward-looking approximations whose coefficients are combinations of the underlying structural parameters. The parameters that are estimated in this paper are therefore reduced form throughout, and I attach no structural interpretation to them beyond their signs. In the trade relation, import and export volumes depend on relative prices and on activity at home and abroad; quantities respond to prices only gradually, such that the real exchange rate enters with a lag.

I depart from these general structures in two key respects. Trend output and the natural rate of exchange are treated as stochastic processes rather than assumed constant, following the unobserved-components tradition and the natural rate of exchange is given an observed component in the terms of trade, so that part of the equilibrium rate is tied to a measured fundamental as in behavioural equilibrium models. The terms of trade are chosen because an improvement in the relative price of exports raises the real exchange rate consistent with any given external position, and because a long quarterly series is available. Net foreign assets would be the natural complement (Lane and Milesi-Ferretti, 2007), but quarterly data for the United States are scarce over much of the sample.

2.2 Measurement equations

Four series are observed: y_t, the log of real output; π_t, core inflation; q_t, the log real effective exchange rate, rising with an appreciation; and n_t, net exports as a share of nominal output. Three further observed variables enter the model exogenously: the terms of trade τ_t, the real rate gap $\tilde{r}_t$ between the ex-ante real rate and the natural rate of interest, and the long-rate differential Δr_t between the United States and a GDP-weighted average of four G7 members whose long-term yields cover the full sample (refer to Section A for data sources).

The measurement equations of the system are presented as follows:

$$y_t = y_t^* + a_{y,1}\tilde{y}_{t-1} + a_{y,2}\tilde{y}_{t-2} + a_q\tilde{q}_{t-1} + a_r\tilde{r}_{t-1} + \varepsilon_t^y, \tag{1}$$

$$\pi_t = b_\pi\pi_{t-1} + (1-b_\pi)\,\bar{\pi}_{t-2,t-4} + b_y\tilde{y}_{t-1} + b_q\tilde{q}_{t-1} + \varepsilon_t^\pi, \tag{2}$$

$$q_t = q_t^* + \rho_{q,1}\tilde{q}_{t-1} + \rho_{q,2}\tilde{q}_{t-2} + c_y\tilde{y}_{t-1} + c_r\Delta r_{t-1} + \varepsilon_t^q, \tag{3}$$

$$n_t = n_{t-1} + \gamma_q\tilde{q}_{t-2} + \gamma_y\tilde{y}_t + \gamma_\tau\Delta\tau_t + \varepsilon_t^n, \tag{4}$$

$$q_t^* = \beta_\tau\tau_t + \mu_t. \tag{5}$$

Equation (1) is the IS curve, written as an equation for output; a_q and a_r are the responses of demand to the real exchange rate gap ($\tilde{q}_t = q_t - q_t^*$) and to the real rate gap ($\tilde{r}_t = r_t - r_t^*$). The output gap ($\tilde{y}_t = y_t - y_t^*$) enters with two lags, selected by both the Akaike and Schwarz information criteria. The real interest rate enters the dynamic IS curve in deviation from its natural rate, sourced directly from Holston et al. (2023).

Equation (2) is an accelerationist Phillips curve in which the coefficients on lagged inflation sum to unity, where $\bar{\pi}_{t-2,t-4}$ represents the average of the second through to fourth lags of inflation. This equation has neither an intercept nor a term in the level of inflation, and thus carries no implication about the long-run inflation rate, which is stable only when both the output gap and the exchange rate gap are zero.

Equation (3) is the interest-parity relation, written as an equation for the real exchange rate; $\rho_{q,1}$ and $\rho_{q,2}$ govern the speed at which it returns to the natural rate and c_r its response to the interest differential. Modelling q_t is what makes $\tilde{q}_t$ a mean-reverting gap rather than an unrestricted residual. With Δr and $\tilde{y}$ constant the gap settles at $(c_r\Delta r + c_y\tilde{y})/(1-\rho_{q,1}-\rho_{q,2})$, such that the natural rate is the rate consistent with internal and external balance at a zero long-run rate differential.

Net exports (4) is specified in first differences, which the data favour over the specification in levels. Augmented Dickey Fuller (ADF) tests fail to reject a unit root in net exports, with or without a trend, KPSS tests reject stationarity around both, and neither the lagged level nor a drift is significant when added to the differenced equation. Furthermore, I also exclude

a latent component from (4), since any component flexible enough to track net exports would also absorb the exchange rate variation the equation is meant to identify. Section 5 reports results for the specification in levels.

As net exports are measured in nominal terms, (4) also includes the change in the terms of trade. A shift in relative prices changes the nominal trade balance before volumes have time to adjust, and without this term the valuation effect would be attributable to the exchange rate gap. The real exchange rate appears in the equation only through the gap $\tilde{q}_{t-2}$. Given movements in q_t^* are by construction consistent with external balance, they do not exert any pressure on net exports, which therefore respond only to deviations of the real exchange rate from its equilibrium.

Equation (5) outlines the law of motion for the natural rate of exchange q_t^*, defined as the real exchange rate at which net exports are neither improving nor deteriorating. This should be read as a condition on the change in the external position, rather than its level, so that the natural rate is not pinned to any particular trade balance and carries no implication drawn from an intertemporal budget constraint nor from a target for net foreign assets. The natural rate of exchange estimated in this paper is accordingly the rate at which net exports are free from exchange rate and cyclical pressure; a weaker condition than the external sustainability requirements often found in fundamental equilibrium exchange rate approaches.

2.3 Transition equations

Trend output is assumed to evolve as a random walk with drift given by trend growth, which in turn follows a random walk, whereas the latent component of the natural rate is modelled as a driftless random walk, consistent with Laubach and Williams (2003):

$$y_t^* = y_{t-1}^* + g_{t-1} + \varepsilon_t^{y^*}, \tag{6}$$

$$g_t = g_{t-1} + \varepsilon_t^g, \tag{7}$$

$$\mu_t = \mu_{t-1} + \varepsilon_t^\mu. \tag{8}$$

All disturbances are Gaussian, mutually independent and serially uncorrelated. Stationarity of the autoregressive terms in equations (1) and (3) implies both gaps are mean-reverting by construction and (6)-(8) carry the permanent components of output and the exchange rate.

Two features of this specification matter for what follows. Independence of the permanent and transitory disturbances is what allows the trend to be separated from the gaps. Variances are constant except over the pandemic period, when the measurement disturbances are scaled to deal with extreme volatility in the data, as outlined in the following section.

2.4 State-space form and the pandemic

The model (1)-(8) is a linear Gaussian state-space system in seven states:

$$\xi_t = \left(y_t^*,\ y_{t-1}^*,\ y_{t-2}^*,\ g_t,\ \mu_t,\ \mu_{t-1},\ \mu_{t-2} \right)'. \tag{9}$$

As the output gap and the real exchange rate gap are each an observed series minus a trend, the state vector need only contain the trends and the lags required to form them. The state vector therefore incorporates two lags of y_t^* and μ_t, as both gaps enter the model up to $t-2$. Each measurement equation is consequently linear in ξ_t, with the observed terms collected in a time-varying intercept, and each equation carrying its own disturbance term. Section B presents the complete state-space representation at each stage of the estimation.

Shock standard deviations are scaled during the pandemic following Lenza and Primiceri (2022): the four measurement disturbances are multiplied by κ_0 in 2020Q2, κ_1 in 2020Q3, and thereafter by $1+(\kappa_1-1)\rho_\kappa^j$, with all three parameters estimated. The trend disturbances in (6)-(8) are left unscaled, so that the contraction and recovery of 2020 are attributed to the gaps rather than to trend output and the natural rate. Section 5 investigates two alternative specifications, including treating 2020Q2-Q4 as missing.

3 Estimation

The state-space system is estimated by maximum likelihood over 1970Q1-2025Q4, with the likelihood being evaluated by the Kalman filter. The trends are initialised at values estimated from the beginning of the sample, and the first four quarters are excluded from the likelihood function for initialisation of the filter.

Two features of the parameter space require attention. Both gaps must be stationary for any decomposition to be meaningful and are therefore parameterised through their partial autocorrelations, making (1) and (3) stationary by construction; (4) has a unit coefficient on n_{t-1} and no intercept by construction. The response of net exports to the gap is restricted by expenditure switching, such that $\gamma_q \leq 0$. In the absence of this restriction the likelihood has a second mode, approximately eight points higher, in which a large positive γ_q lets a small, transitory gap fit the noise in net exports with minimal misalignment. The likelihood is also multimodal in the trend/cycle division, so estimation starts from a grid of initial values, and each alternative specification is also started from the baseline estimates.

Estimation follows the three-stage procedure of Laubach and Williams (2003) in which the median-unbiased estimator is applied in stages to avoid the pile-up problem (Stock and Watson, 1998). Stage I first imposes a constant trend growth, $\sigma_g = 0$, and recovers the ratio

$\lambda_g = \sigma_g/\sigma_{y^*}$ from the median-unbiased estimator applied to the smoothed path of the trend growth rate. Stage II imposes λ_g, holds the latent component of the natural rate constant, and recovers $\lambda_q = \sigma_\mu/\sigma_q$ from a test for a shift in the intercept of (3). Stage III then imposes both ratios and estimates the remaining parameters.

I report both one-sided $E[\xi_t|\Omega_t]$, and two-sided $E[\xi_t|\Omega_T]$ estimates, where Ω_t is the information set at t. The reported intervals combine both filter uncertainty about the states with uncertainty about the parameters. The latter is propagated by drawing from the asymptotic distribution of the estimates in the transformed space and re-running the filter and smoother for each draw (Hamilton, 1986). Uncertainty of the imposed ratios λ_g and λ_q is not included, so the intervals are conditional upon both.

3.1 Identification of the trend/cycle division

Two shock variances, those of trend growth and of the latent component of the natural rate, are only weakly determined by the likelihood, which is a well-known feature of unobserved-components models in such applications (Laubach and Williams, 2003). For the trend growth rate, the ratio $\lambda_g = \sigma_g/\sigma_{y^*}$ is estimated in Stage I by the median-unbiased estimator of Stock and Watson (1998) and then held fixed. The test underlying that stage is informative, and the resulting ratio, $\lambda_g = 0.068$, is reported in Table 2.

The variance of the natural rate requires a separate restriction. Net of the terms-of-trade component, the real exchange rate is the sum of the latent trend and the gap by construction, so that with $\varphi_t \equiv q_t - \beta_\tau \tau_t = \mu_t + \tilde{q}_t$,

$$\text{var}(\Delta\varphi_t) = \sigma_\mu^2 + \text{var}(\Delta\tilde{q}_t), \tag{10}$$

where $\text{var}(\Delta\tilde{q}_t)$ is driven mainly by the gap's own disturbances and so is of the order of σ_q^2. The likelihood pins down the sum sharply, but how it divides between the two depends on the dynamics of the gap and on how the gap enters the other equations. With two lags in (3) and the valuation term in (4), the likelihood carries some, albeit limited, information about this division. Leaving both ratios unrestricted, the profile of the likelihood over σ_μ peaks at roughly 0.75; a natural rate that absorbs almost all of the variance in (10) is rejected, but an invariant natural rate is not. The likelihood therefore bounds the volatility of the natural rate from above without determining it precisely.

This volatility is instead fixed through the signal-to-noise ratio $\lambda_q = \sigma_\mu/\sigma_q$, which governs how much of (10) is permanent. The ratio is estimated in Stage II. With the latent component of q^* held constant, equation (3) implies a constant intercept, and the exponential Wald test of Andrews and Ploberger (1994) for a shift in that intercept at an unknown date is mapped

into a median-unbiased estimate of $\lambda_q = 0.39$. Taken on its own, the test has little power to discriminate between values of λ_q. A shift in μ enters (3) scaled by $1 - \rho_{q,1} - \rho_{q,2} \approx 0.090$, so the signal available to the test is small whatever the value of λ_q, and its 90 percent interval extends to the upper limit of the values considered.

Median-unbiased and maximum likelihood estimates of λ_q are nevertheless almost identical. Estimating λ_q jointly with the other parameters of the model by maximum likelihood yields 0.38 (Panel B of Table 3), almost identical to the Stage II estimate, so imposing the latter costs essentially nothing in fit. At the upper end of the Stage II interval the fit deteriorates noticeably, whilst at the lower end it changes minimally. The data therefore rule out a highly volatile natural rate but cannot by themselves distinguish the moderately variable latent component μ_t from a constant, in which case q_t^* moves strictly with the terms of trade.

4 Results

4.1 Parameter estimates

Estimates are reported in Table 2. The real exchange rate gap is persistent but stationary, with estimates of $\rho_{q,1}$ and $\rho_{q,2}$ implying a half-life of roughly two years for a disturbance to the gap. Misalignment itself also inherits the persistence of the interest rate differential (Section 4.2), so this is not directly comparable with the three-to-five year range for deviations from purchasing power parity widely cited in the literature (Rogoff, 1996).

The main open-economy relations in the semi-structural state-space model are estimated precisely and correctly signed in line with standard theory. In particular, an improvement in the terms of trade component appreciates the natural rate of exchange, with a coefficient of 0.59. A widening of the differential between the United States and foreign long-term yields appreciates the real exchange rate relative to its natural rate, consistent with the interest-parity relationship in (3). The implied long-run effect, $c_r/(1 - \rho_{q,1} - \rho_{q,2})$, is 7.3 percent per percentage point, of the order that real interest parity on ten-year bonds would imply. The real rate gap enters the IS curve negatively, and an improvement in the terms of trade raises nominal net exports on impact ($\gamma_\tau = 0.081$).

The real exchange rate gap has only small and imprecisely estimated effects on demand and prices, as might be expected given the modest role of trade in the United States economy. The positive estimate of a_q arguably reflects the tendency of the dollar to appreciate when domestic demand is strong. In net exports, where expenditure switching should be clearly visible, the effect has the expected negative sign and is significant once valuation effects are accounted for. As the effects on output and inflation are small, the estimated gap is largely

insensitive to them. Estimates of misalignment therefore do not hinge on coefficients the data determine only loosely, with the exchange rate equation providing most of the information about the split between the natural rate and the gap.

The remaining parameter estimates are largely conventional. Trend growth of potential output declines from 3.2 percent to 2.2, and the slope coefficient of the Phillips curve is small. These results are consistent with closed-economy estimates of this kind, including those of Holston et al. (2017), suggesting that the open-economy extension leaves the domestic block of the state-space model largely intact.

4.2 The natural rate of exchange

Figure 1 plots two-sided estimates of the natural rate against the actual real exchange rate. The natural rate falls steeply through the 1970s and then once again during the late 1980s, drifts within a narrow range until its low in 2008Q3, and then increases by 19 percent towards the end of the sample. In the model, the observed terms of trade accounts for 55 percent of the variance of the quarterly change in q_t^*, with the latent trend accounting for the remainder. In levels, the two components vary by similar amounts, with standard deviations of 5.6 and 4.4 percent respectively. The observed component peaks just prior to the first oil shock and reaches its low during the commodity boom.

Table 1 reports filtered and smoothed estimates of misalignment at selected dates. Deviations of the actual exchange rate from the natural rate of exchange are persistent. Misalignments hold the same sign for 4 years on average (16 quarters), so the dollar experiences long stretches on one side of its natural rate. Given (3) is linear, the gap can be divided into three components; the differential, the output gap, and accumulated disturbances. The differential accounts for 64 percent of the variance of the gap, whereas the output gap contribution is negligible. A sizeable part of the measured misalignment is therefore reducible to the dollar moving in line with interest rate parity.

Figure 2 plots the time-varying gap between the actual and natural rates of exchange. Periods of large misalignment coincide with well-known historical movements in the dollar. For example, the dollar is well below its natural rate in the late 1970s, with the gap reaching around 20 percent in 1978Q4, during the dollar crisis. It then increases sharply to 20 percent above the natural rate in 1985Q1, shortly before the Plaza Accord. The dollar is once again undervalued in the early 1990s, by 12.6 percent, before the shift to a strong-dollar policy. By 2002 it is 12.1 percent above its natural rate, at a time when the sustainability of the United States external position was widely debated (Obstfeld and Rogoff, 2005). It is overvalued again through the 2020s and ends the sample 8.5 percent above its natural rate.

4.3 Filtered and smoothed estimates

The difference between one-sided and two-sided estimates reported in Table 1 indicates how much the estimate for a given quarter changes once later data are taken into account. These differences are small relative to the gap and concentrated in the first few quarters, where the Kalman filter has little history to draw upon, particularly around turning points. The 1985 peak is a clear example of this outside the initial quarters. Using data up to that quarter, the dollar is well above its natural rate, albeit the size of the overvaluation is understated by more than a third. The terms-of-trade contributes to this stability, since only the latent trend changes, and the estimates are noticeably less stable without it. Such one-sided estimates use current-vintage data and full-sample parameters, so they reflect the end-of-sample problem, which Orphanides and van Norden (2002) find to be the main source of unreliability in the analogous case of the output gap, but not the effects of data revisions or re-estimation.

4.4 Out-of-sample forecasts

If the natural rate of exchange is a meaningful measure, departures from it should be followed by movements of the real exchange rate back to its equilibrium. A common test of equilibrium exchange rate models, as in Ca' Zorzi et al. (2022), is whether misalignment can forecast the real exchange rate better than a random walk, a benchmark that has long proven difficult to outperform since Meese and Rogoff (1983). I apply this test to the natural rate of exchange from 1990Q1 to 2025Q4, using data prior to this as the initial estimation sample.

At each quarter from 1990Q1, estimates use only data available up to that quarter. The parameters are re-estimated in the fourth quarter of every year, with ratios λ_g and λ_q held at their baseline values, and the one-sided gap at each quarter is used to forecast the change in the real exchange rate over the following one to four years. These forecasts follow directly from the model: the natural rate is assumed to remain at its current level, and the exchange rate gap evolves according to equation (3) with the interest differential fixed at its current value and the output gap omitted. The forecast change in the real exchange rate is therefore the expected change in the exchange rate misalignment gap.

The forecasts in this exercise are compared with a random walk, which predicts no change, and with two statistical measures of misalignment, namely the deviation of the real exchange rate from its mean up to each quarter, as is implied by purchasing power parity, and a one-sided Hodrick–Prescott gap. In each case, the forecast is obtained by regressing the change in the real exchange rate on the gap, using only data available at the time of the forecast. Two restricted versions of the model forecast isolate its components, one setting the interest differential to zero and the other setting the gap to zero.

Panel C of Table 3 reports the main results. The natural rate forecasts the real exchange rate better than a random walk at every horizon, with a MSE that is 9 percent lower at the one year horizon and 31 percent lower at a four year horizon. The improvement is significant at three and four years in the Diebold and Mariano (1995) test, and at every horizon in the Clark and West (2007) test, which accounts for a nested random walk. The forecast predicts the direction of change correctly in around two-thirds of quarters. Neither statistical measure improves significantly on the random walk as much as the natural rate of exchange.

5 Robustness

To investigate the robustness of these results, Table 3 reports a range of alternative specifications. Omitting the pandemic period or the volatility scaling altogether leaves the persistence of the exchange rate gap and the volatility of the natural rate almost unchanged. Dropping the terms of trade or the interest differential worsens the fit significantly, while writing (4) in levels has a negligible impact. Weighting foreign yields by trade raises the response of the exchange rate to the differential but leaves misalignment almost unchanged. Estimates on a sample ending prior to the pandemic in 2019Q4, with λ_g re-estimated and λ_q held fixed at its full-sample value, are also close to the baseline.

Three further checks concern the components of the model. The first relates to the terms of trade. Since they are defined as a ratio of dollar deflators, the exchange value of the dollar itself may influence them. Entering the terms of trade with a lag removes that channel. The associated coefficient then declines essentially to zero and the fit worsens markedly, although the broad path of misalignment is preserved. The terms-of-trade component is therefore best read as a contemporaneous relationship rather than as the causal effect of a fundamental. It nonetheless remains the only observed fundamental the data support, as adding net foreign assets, measured by the cumulated current account, or the effective tariff rate to the natural rate leaves the fit almost unchanged (Panel A of Table 3).

The remaining two robustness checks concern features that distinguish the baseline estimate from a simpler specification, namely the second lag in equation (3) and the valuation term in equation (4). I find both are supported by the data. With a single lag in the former the fit is worse and the residuals are autocorrelated, which would thus invalidate the white-noise assumption underlying the Stage II test. Dropping the valuation term from the latter also worsens the fit considerably. Removing both, however, leaves the path of misalignment almost unchanged. Their contribution thus lies instead in how precisely the data determine the wedge between the natural rate of exchange and the gap.

6 Conclusion

This paper measures the natural rate of exchange for the United States from 1970Q1–2025Q4 by estimating a semi-structural state-space open economy model with maximum likelihood and identifying the trend/cycle division by way of median-unbiased estimation. The natural rate is defined as the real effective exchange rate consistent with output at potential, stable inflation and a stable trade balance, and is modelled as an observed terms-of-trade component in addition to a latent stochastic trend.

Baseline estimates show that the natural rate of exchange changes considerably over time, falling for much of the sample before rising after the financial crisis of 2008, with the terms of trade accounting for a large share of its short-run variation. Departures of the dollar from its natural rate are both large and long-lived, and much of their magnitude and persistence can be traced to the long-term interest rate differential. One-sided Kalman filtered estimates are close to their two-sided smoothed counterparts, and the historical episodes of misalignment are robust across a range of model specifications. Out of sample, the natural rate of exchange forecasts the dollar better than a random walk at horizons of one to four years, which neither purchasing power parity nor a statistical trend achieves.

These findings suggest that judgements about dollar misalignment are substantially dependent on allowing the equilibrium to vary over time. Invariant measures instead attribute low-frequency movements in the real exchange rate to misalignment alone, whereas a purely statistical trend absorbs much of what is better seen as persistent departures from equilibrium. Treating the natural rate as a latent state, disciplined by the terms of trade alongside the relationships that define it, places the estimates between these extremes.

As with other semi-structural estimates of unobserved natural rates, these findings rest on several modelling choices. The natural rate of exchange is defined by internal and external balance, but the division between the natural rate and the real exchange rate gap draws on the exchange rate equation, and the estimates are conditional on the imposed ratio λ_q. The broad timing and direction of misalignment are largely robust to these choices.

These choices also point to several natural extensions. A full account of uncertainty would require a simulation study of the three-stage procedure, together with intervals that integrate over λ_q. The model itself is open to further adjustment. Measuring the interest differential with alternative inflation expectations might sharpen the interest-parity interpretation of the gap, and as quarterly data on net foreign assets lengthen, they could also enter the natural rate alongside the terms of trade. Finally, applying the framework to other currencies could show whether our results are specific to the dollar. Such extensions are left to future research.

Table 1: Exchange rate misalignment

	Misalignment $q - q^*$			
	One-sided		Two-sided	
Quarter	Est.	90% band	Est.	90% band
1970Q1	−7.6	[−22.3, 9.1]	−3.2	[−10.5, 7.0]
1978Q4	−19.4	[−26.1, −11.5]	−20.0	[−24.0, −14.0]
1985Q1	11.9	[3.8, 21.2]	19.8	[13.3, 27.0]
1992Q3	−13.0	[−19.7, −6.8]	−12.6	[−17.5, −6.9]
2002Q1	8.2	[0.1, 16.2]	12.1	[5.4, 17.5]
2011Q2	−7.3	[−14.3, −1.5]	−9.7	[−15.1, −4.9]
2020Q2	18.6	[9.5, 26.6]	16.7	[8.4, 24.1]
2021Q2	7.3	[−1.6, 14.7]	4.9	[−2.5, 11.8]
2025Q4	8.5	[−0.1, 17.7]	8.5	[−0.3, 17.2]

Notes: Estimates of misalignment between the real effective exchange rate and its natural rate. q is defined as 100 log of the BIS real narrow effective exchange rate. One-sided estimates use information up to t; two-sided estimates use the full sample. Misalignment is the percentage deviation of the real exchange rate from its natural rate, positive when the dollar is above its natural rate. Bands are 90 percent and combine filter with parameter uncertainty, conditional on the imposed ratios λ_g and λ_q of trend growth and the natural rate respectively. Dates are the first and last quarters of the sample and the turning points of the two-sided estimate, dated by a version of the Harding and Pagan (2002) rule with a three-year window.

Table 2: Parameter estimates

		Estimate	(s.e.)
IS curve			
$a_{y,1}$	Output gap, first	1.212	(0.098)
$a_{y,2}$	Output gap, second	-0.257	(0.100)
a_q	Exchange rate gap	0.010	(0.006)
a_r	Interest rate gap	-0.068	(0.022)
σ_y	Disturbance s.d.	0.644	(0.061)
Phillips curve			
b_π	Inflation	0.706	(0.065)
b_y	Output gap	0.049	(0.027)
b_q	Exchange rate gap	-0.010	(0.007)
σ_π	Disturbance s.d.	0.863	(0.041)
Real exchange rate			
$\rho_{q,1}$	Exchange rate gap, first	1.203	(0.076)
$\rho_{q,2}$	Exchange rate gap, second	-0.293	(0.070)
c_y	Output gap	-0.008	(0.080)
c_r	Interest rate differential	0.654	(0.226)
σ_q	Disturbance s.d.	2.070	(0.100)
Natural rate of exchange			
β_τ	Terms of trade	0.588	(0.110)
σ_μ	Innovation s.d.	0.813	(–)
Net exports			
γ_q	Exchange rate gap	-0.006	(0.003)
γ_y	Output gap	0.018	(0.010)
γ_τ	Change in terms of trade	0.081	(0.014)
σ_n	Disturbance s.d.	0.289	(0.014)
Trend output			
σ_{y^*}	Innovation s.d., level	0.304	(0.097)
σ_g	Innovation s.d., growth	0.021	(–)
Volatility scale			
κ_0	2020Q2	6.780	(2.478)
κ_1	2020Q3	7.050	(2.655)
ρ_κ	Decay	0.051	(0.141)

Notes: Final stage maximum likelihood estimates from 1970Q1 to 2025Q4. Dashes mark imposed parameters, with σ_μ implied by the median-unbiased ratio $\lambda_q = 0.39$ and σ_g by $\lambda_g = 0.068$. Standard errors are computed from the numerical Hessian of the log-likelihood and mapped to the natural parameters by the delta method.

Table 3: Robustness and performance

	$\log L$	$\Sigma_j \rho_{q,j}$	σ_μ	$\sigma_{\tilde{q}}$
Panel A. Specification				
Baseline	−1071.8	0.910	0.81	9.1
2020Q2–Q4 omitted	−1034.9	0.891	0.81	8.6
No volatility scaling	−1154.4	0.910	0.83	9.4
No terms of trade, $\beta_\tau = 0$	−1084.8	0.945	0.88	10.3
No interest differential, $c_r = 0$	−1076.6	0.971	0.84	7.7
Net exports in levels, quadratic trend	−1066.8	0.910	0.81	9.1
Terms of trade lagged, τ_{t-1}	−1084.8	0.946	0.88	10.4
Net foreign assets in q^*	−1071.7	0.909	0.81	9.3
Tariff rate in q^*	−1070.9	0.913	0.81	9.0
Foreign yields weighted by trade	−1071.0	0.904	0.81	9.2
AR(1) exchange rate gap, $\rho_{q,2} = 0$	−1079.7	0.909	0.84	8.9
No $\Delta\tau_t$ in net exports, $\gamma_\tau = 0$	−1088.2	0.911	0.82	9.2
AR(1) gap and $\gamma_\tau = 0$	−1096.2	0.911	0.85	9.0
Panel B. Imposed ratio				
Estimated jointly	−1071.8	0.910	0.78	9.1
$\lambda_q = 0.00$, lower 90% limit	−1072.9	0.930	0.00	9.0
$\lambda_q = 0.39$, baseline value	−1071.8	0.910	0.81	9.1
$\lambda_q = 3.00$, upper 90% limit	−1075.2	0.932	2.14	6.2
Forecast horizon, h quarters:	4	8	12	16
Panel C. Out-of-sample forecasts				
Natural rate of exchange	0.91	0.81	0.73*	0.69**
No differential, $\Delta r_t = 0$	1.33	1.32	1.20	1.00
No misalignment, $\tilde{q}_t = 0$	1.04	1.06	1.05	1.04
Purchasing power parity	0.99	0.97	0.98	0.94
Hodrick-Prescott filter	1.12	1.13	1.05	1.01

Notes: Panels A and B report the log-likelihood, the sum of the autoregressive coefficients on the exchange rate gap, the standard deviations of innovations to the natural rate and of the two-sided gap. Panel A holds λ_g and λ_q at their baseline values; Panel B varies λ_q across its 90 percent interval. Panel C reports the mean squared error of forecasts of the change in the real exchange rate over h quarters from 1990Q1, relative to a random walk, with the model estimated on data up to each quarter. Asterisks mark a significantly smaller mean squared error than the random walk in the Diebold–Mariano test at the 10 (*) and 5 (**) percent.

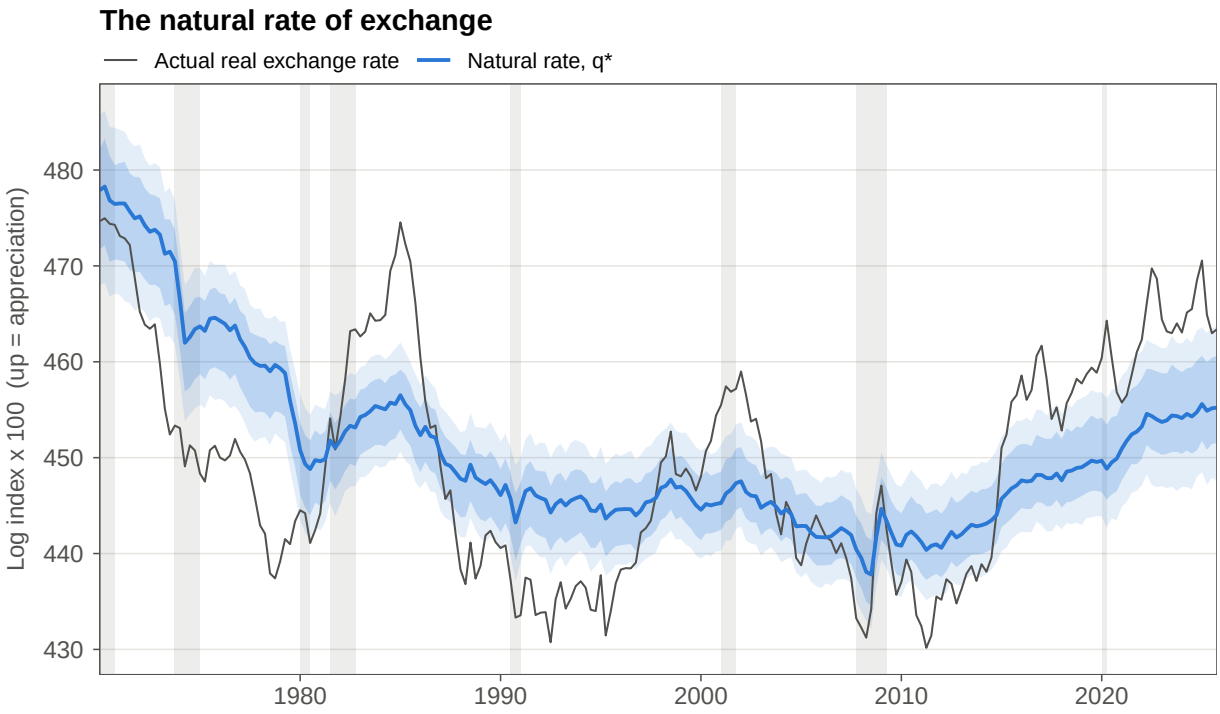


Figure 1: The natural rate of exchange

Notes: Two-sided estimate of the natural rate q^* against the real effective exchange rate q. Both time-series are 100 times the log of the index, 2020 = 100. Shaded bands are the 68 and 90 percent intervals combining filter and parameter uncertainty. Grey bars mark recessions from the National Bureau of Economic Research.

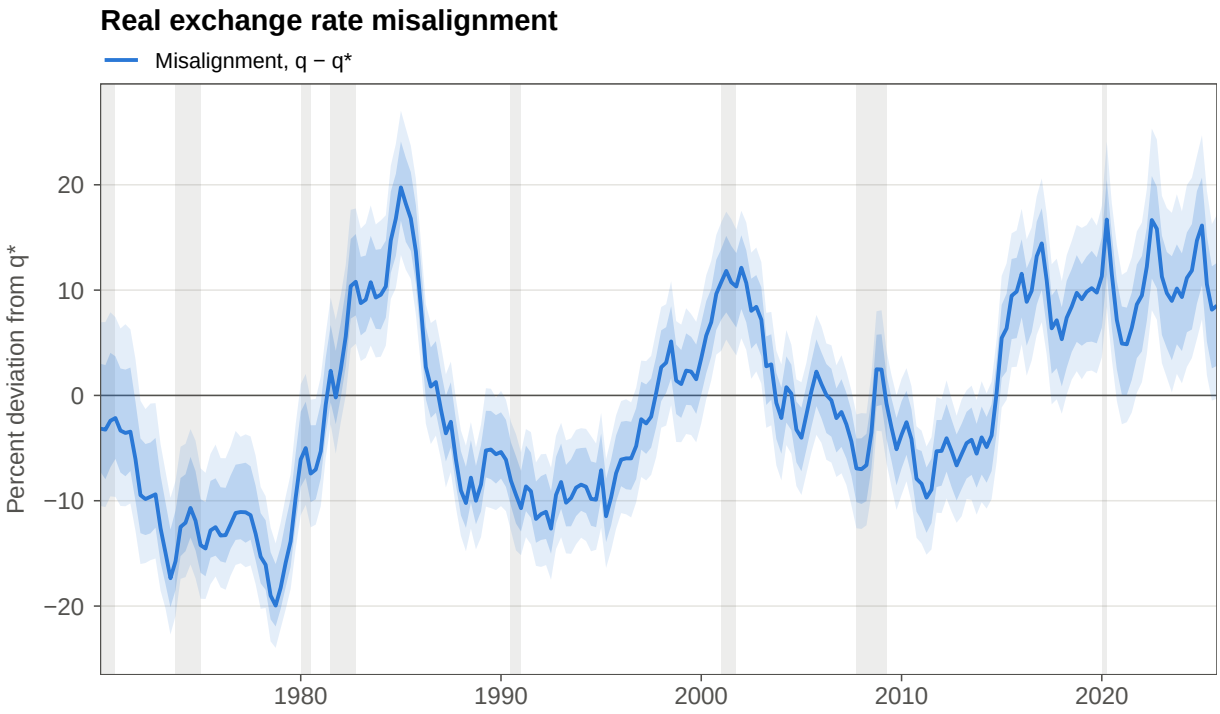


Figure 2: Dollar misalignment

Notes: Smoothed estimate of misalignment, $q_t - q_t^*$, measured as the percentage deviation of the real exchange rate from its natural rate. Positive values indicate a real exchange rate above its natural rate (overvaluation) and negative values a rate below it (undervaluation).

A Data

The sample is 1970Q1–2025Q4, all series are quarterly. Output is real gross domestic product (FRED `GDPC1`), and entered as 100 log. Inflation is the annualised quarterly log change in the core personal consumption expenditures index (`PCEPILFE`), $400\,\Delta$ log. The real exchange rate is the Bank for International Settlements real narrow effective exchange rate for the United States (`RNUSBIS`), entered as 100 log and oriented so that an increase is an appreciation. Net exports are nominal net exports of goods and services (`NETEXP`) as a percent of nominal gross domestic product (`GDP`). The terms of trade, τ_t, are 100 log of the ratio of the implicit price deflator for exports (`A020RD3Q086SBEA`) to that for imports (`A021RD3Q086SBEA`). The real rate gap is the federal funds rate, less a four-quarter moving average of core PCE inflation, less the one-sided estimate of the natural rate of interest, all three as published by the Federal Reserve Bank of New York for Holston et al. (2023). The interest rate differential is the real ten-year Treasury yield less a weighted average of the real long-term government bond yields of Canada, France, Germany and the United Kingdom (`IRLTLT01{CA,FR,DE,GB}Q156N`), the G7 economies whose series cover the complete sample; Japan and Italy begin only in 1989 and 1991. The weights are each economy's total share of the four economies' GDP in current dollars (World Bank), averaged from 1970 and held fixed. All monthly series are averaged to quarterly frequency. As q_t is an index, the level of q_t^* inherits its normalisation (2020 = 100); misalignment $q_t - q_t^*$ is invariant to it and is the object I interpret, reported throughout as the exact percentage deviation of the real exchange rate from its natural rate.

B Model specification

The system is estimated in three stages, in the state-space notation of Hamilton (1994).

The observation and state equations are

$$\mathbf{y}_t = \mathbf{A}'\mathbf{x}_t + \mathbf{H}'\boldsymbol{\xi}_t + \mathbf{w}_t, \qquad \boldsymbol{\xi}_t = \mathbf{F}\boldsymbol{\xi}_{t-1} + \mathbf{v}_t, \qquad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_t), \;\; \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t), \tag{11}$$

where $\mathbf{x}_t$ collects the predetermined and exogenous variables:

$$\begin{aligned}
\mathbf{y}_t &= (\, y_t,\; \pi_t,\; q_t,\; n_t \,)', \\
\mathbf{x}_t &= (\, y_t,\; y_{t-1},\; y_{t-2},\; \pi_{t-1},\; \bar{\pi}_{t-2,t-4},\; q_{t-1},\; q_{t-2},\; \tau_t,\; \tau_{t-1},\; \tau_{t-2},\; \tilde{r}_{t-1},\; \Delta r_{t-1},\; n_{t-1},\; \Delta\tau_t \,)', \\
\boldsymbol{\xi}_t &= (\, y^*_t,\; y^*_{t-1},\; y^*_{t-2},\; g_t,\; \mu_t,\; \mu_{t-1},\; \mu_{t-2} \,)'.
\end{aligned}$$

Current output y_t enters the vector $\mathbf{x}_t$ only through the net-exports equation, whose output gap is contemporaneous. As every gap in this system is an observable less a state, $\tilde{y}_t = y_t - y^*_t$ and $\tilde{q}_t = q_t - \beta_\tau\,\tau_t - \mu_t$, the loadings are

$$\mathbf{H}' = \begin{bmatrix}
1 & -a_{y,1} & -a_{y,2} & 0 & 0 & -a_q & 0 \\
0 & -b_y & 0 & 0 & 0 & -b_q & 0 \\
0 & -c_y & 0 & 0 & 1 & -\rho_{q,1} & -\rho_{q,2} \\
-\gamma_y & 0 & 0 & 0 & 0 & 0 & -\gamma_q
\end{bmatrix}, \qquad \mathbf{F} = \begin{bmatrix}
1 & 0 & 0 & 1 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0
\end{bmatrix},$$

and the rows of $\mathbf{A}'$ collect the corresponding observable terms, products of parameters included. The covariance matrices are diagonal, specified as follows:

$$\begin{aligned}
\operatorname{diag}(\mathbf{Q}) &= \big(\sigma^2_{y^*},\;\; 0,\;\; 0,\;\; \sigma^2_g,\;\; \sigma^2_\mu,\;\; 0,\;\; 0\big), \\
\operatorname{diag}(\mathbf{R}_t) &= \kappa^2_t\big(\sigma^2_y,\;\; \sigma^2_\pi,\;\; \sigma^2_q,\;\; \sigma^2_n\big),
\end{aligned}$$

where κ_t is the volatility scale of Section 2.4, equal to one before 2020Q2. Every measurement equation carries its own disturbance, so $\mathbf{R}_t$ is non-singular; the zeros in $\operatorname{diag}(\mathbf{Q})$ correspond to the lag-shift states, which carry no independent disturbance.

B.1 Multi-stage maximum likelihood

Stage I. Imposes $\sigma_g = 0$, so that the trend growth rate is constant and g is a parameter rather than a state. The initial vector estimated by maximum likelihood is therefore:

$$\theta_{\mathrm{I}} = \big(\, a_{y,1},\ a_{y,2},\ a_q,\ a_r,\ b_\pi,\ b_y,\ b_q,\ \rho_{q,1},\ \rho_{q,2},\ c_y,\ c_r,\ \beta_\tau,\ \gamma_q,\ \gamma_y,\ \gamma_\tau,\\ g,\ \sigma_y,\ \sigma_\pi,\ \sigma_q,\ \sigma_n,\ \sigma_{y^*},\ \sigma_\mu,\ \kappa_0,\ \kappa_1,\ \rho_\kappa \,\big).$$

The ratio $\lambda_g = \sigma_g/\sigma_{y^*}$ follows from the Stock and Watson (1998) median-unbiased estimator applied to the structural-break statistic of the smoothed potential-output growth path $\Delta\hat{y}^*_{t|T}$.

Stage II. Imposes $\sigma_g = \hat{\lambda}_g\sigma_{y^*}$, so that g becomes a state, and holds the latent component of the natural rate constant, $\sigma_\mu = 0$. The remaining parameters other than $\rho_{q,1}$ and $\rho_{q,2}$ are estimated by maximum likelihood. With μ constant, equation (3) implies that

$$z_t \equiv \varphi_t - \rho_{q,1}\varphi_{t-1} - \rho_{q,2}\varphi_{t-2} - c_y\tilde{y}_{t-1} - c_r\Delta r_{t-1} = (1-\rho_{q,1}-\rho_{q,2})\,\mu + \varepsilon^q_t, \qquad \varphi_t \equiv q_t - \beta_\tau\,\tau_t,$$

has a constant mean. The exponential Wald statistic for a shift in that mean at an unknown date gives $\lambda_q = \sigma_\mu/\sigma_q$ through the median-unbiased estimator, with the mapping from the ratio to the statistic simulated for the exchange rate equation. Because that mapping depends on $(\rho_{q,1}, \rho_{q,2})$, and their Stage III estimates depend on λ_q, they are held at their Stage III values and the two stages are iterated to a fixed point.

Stage III. Imposes both ratios, so that the fourth and fifth elements of $\mathrm{diag}(\mathbf{Q})$ are $\hat{\lambda}^2_g\sigma^2_{y^*}$ and $\hat{\lambda}^2_q\sigma^2_q$. The vector estimated by maximum likelihood is $\theta_{\mathrm{III}} = \theta_{\mathrm{I}} \setminus \{g, \sigma_\mu\}$, of dimension 23. One- and two-sided state estimates follow from the filter and the smoother at $\hat{\theta}_{\mathrm{III}}$, and the reported intervals additionally integrate over its asymptotic distribution.

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